GeoNatureAgent Benchmark: Benchmarking LLM Agents for Environmental Geospatial Analysis Across Frontier and Open-Weight Foundation Models [Benchmark]

Gabriel Diaz-Ireland*
gdiazir1@jh.edu
Universidad Católica de Ávila (UCAV)
Ávila, Spain
Johns Hopkins University
Baltimore, MD, USA

Diego Prieto-Herráez
diego.prieto@ucavila.es
Universidad Católica de Ávila (UCAV)
Ávila, Spain

Mario García Peces
mario.garcia.peces@gmail.com
Independent Researcher
Madrid, Spain

Javier Velázquez
javier.velazquez@ucavila.es
Universidad Católica de Ávila (UCAV)
Ávila, Spain

Devika Jain
kakkar@fas.harvard.edu
Center for Geographic Analysis,
Harvard University
Cambridge, MA, USA

Abstract

Environmental scientists spend disproportionate effort on data wrangling rather than analysis. New AI agents can be a helpful tool, but their use still requires validation. Furthermore, no benchmark exists to evaluate AI agents that automate environmental geospatial workflows through structured tool calling against real APIs. We introduce the **GeoNatureAgent Benchmark**, the first benchmark for environmental analysis agents that operate via structured tool calls to a production-style geospatial API. The benchmark comprises 93 tasks across 18 categories, covering municipality-level analysis, multi-turn conversation, spatial reasoning, cross-indicator synthesis, error handling and recovery, ranking, comparison, multilingual understanding, habitat analysis, deep-dive profiling, temporal change, and task rejection. Tasks are evaluated against an open, self-hostable geospatial API that serves three environmental indicators across Spain and Portugal via sixteen tools (twelve domain-specific operations and four auxiliary). We evaluate seven large language models (Claude Sonnet 4, DeepSeek V3.2, GLM-5, Gemini 2.5 Pro, Qwen3-235B, GPT-OSS-120B, and Llama 4 Scout) across Vertex AI and Anthropic platforms under three temperature-1.0 seeds per model, reporting capability and per-case cost as orthogonal axes. Results manifest that: (1) Claude Sonnet 4 achieves the highest capability at 60.8%±0.8%, followed closely by DeepSeek V3.2 at 56.3% ± 3.1%, while no other model exceeds 51%; (2) the cost-accuracy Pareto frontier is occupied mostly by open-weight models (Llama 4 Scout → Qwen3-235B → DeepSeek V3.2 → Claude Sonnet 4), with DeepSeek V3.2 offering 93% of Claude's capability at 11× lower cost($0.011/case); (3) comparison tasks remain universally unsolved (0% on most models for close-value comparisons), exposing systematic reasoning limitations; and (4) structured tool calling against a real API provides a more discriminative measure of real-world agent capability, with mean accuracies 25–35 percentage points below those reported on general-purpose GIS benchmarks. We further demonstrate geographic and domain extensibility by integrating BigEarthNet V2 land-cover data for Portugal alongside Spanish CO_2 suitability and gully erosion indicators. GeoNatureAgent Benchmark, the evaluation harness, and the self-hostable API are publicly available[1].

CCS Concepts

• **Computing methodologies → Neural networks**; *Intelligent agents*; *Multi-agent systems*; *Information extraction*; • **General and reference → Evaluation**; • **Information systems → Geographic information systems**; • **Applied computing** → *Earth and atmospheric sciences.*

Keywords

Geospatial AI, Benchmark, LLM Agents, Tool Calling, Environmental Analysis

*Corresponding author.

[1]https://github.com/gabrielireland/GeoNatureAgent_Benchmark